

# PEP PERFORMANCE REPORT

## Flow Analysis | Data Pipeline Methodology

|                     |                                                                         |                       |                                                                     |
|---------------------|-------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------|
| <b>Date:</b>        | 20 May 2026                                                             | <b>Project:</b>       | Global Events, Commodities and Currency Markets - DSB Group Project |
| <b>Prepared By:</b> | Tabassum Nadavinmani (Team Leader) & Abdulhadi Kirkici (Co-Team Leader) | <b>Report Period:</b> | April 2026 – May 2026                                               |

### 1. Summary

This Performance Report summarizes the status, progress, and key performance indicators (KPIs) for the DSB Group Project. The project leverages a Data Pipeline methodology to analyse five datasets: Brent Oil (1946–2026), Gold (1915–2026), Silver (1915–2026), the SWIFT Global Currency Tracker (January–April 2026), and USDG/Global Dollar stablecoin (April 2025–April 2026). The analysis covers the current geopolitical conflict between the United States and Iran, the 2026 Hormuz shipping crisis, energy supply shocks, central bank gold accumulation, and BRICS bloc involvement in non-dollar settlement systems.

As for the report date, all Part I data analytics (30 questions + 6 bonus) are complete, and all 11 charts have been generated and saved. Part II market research (16 questions) has not yet started; it is scheduled to begin the week of 2026-05-22 per the project plan. The project is currently completing Sprint 2 (Part I analysis) and preparing to move into Sprint 3 (Part II market research, followed by final report writing). The report covers project scope adherence, schedule, resource utilisation, deliverables, risks, and additional relevant areas.

### 2. Project Scope and Objectives

#### Scope

Analyse and compare commodity price trends and global currency indicators using a Data Pipeline methodology, connecting observed data patterns to major global events including the Hormuz crisis, energy supply disruptions, gold accumulation, sanctions, BRICS settlement discussions, and currency market shifts.

Datasets in scope:

- Brent Oil price history - 963 monthly observations, 1946–2026
- Gold price history - 1,336 monthly observations, 1915–2026
- Silver price history - 1,336 monthly observations, 1915–2026
- SWIFT Global Currency Tracker - 330 rows, 8 metrics, Jan–Apr 2026
- USDG (Global Dollar stablecoin) - 366 daily observations, Apr 2025–Apr 2026 (collected via CoinGecko API)

#### Objectives

- Collect and integrate all relevant datasets into a standardised, reproducible pipeline.
- Cleanse, transform, and validate data to ensure analytical integrity.
- Generate descriptive, exploratory, predictive, and prescriptive analytics across four analytical layers.
- Produce comparative visualizations linking commodity and currency movements to global events.
- Deliver actionable, evidence-based insights written for non-technical stakeholders.

### 3. Performance Analysis

#### A. Work Completed This Period

- Defined project roles and responsibilities for all five team members.
- Created an organized project folder structure, a Trello board, and a WhatsApp communication group.
- Finalised Data Pipeline architecture: Collect -> Clean -> Transform -> Analyze -> Visualize -> Report.
- Extracted and cleaned all five datasets; resolved USDG duplicate date and documented SWIFT structural NaN values.
- Collected USDG (Global Dollar stablecoin) data independently via CoinGecko public API; documented methodology for reproducibility and academic integrity.
- Conducted full descriptive analytics: date ranges, row counts, data quality checks, feature engineering.
- Completed all 30 Part 1 questions across sections A–E (Data Preparation, Descriptive Analytics, Trend & Visual Analytics, Currency Tracker Analytics, Simple Comparison).
- Completed all 6 optional Bonus analytics (B1–B6): percentage change, correlation table, heatmap, scatter plot, moving average, forecast trendline.
- Produced 11 final output charts, including time series line charts for Brent Oil, Gold, and Silver; a combined Gold–Silver trend chart and a 10-year Brent annual-average bar chart; a correlation heatmap, a Gold–Silver scatter plot, and a Gold moving-average chart; a Gold price forecast chart, a top-10 SWIFT currency bar chart, and a summary dashboard.

#### B. Work Planned for Next Period

- Complete Part II Market Research (16 questions, Sections A–D: Event Identification, Commodity Linking, Currency Linking, Business Interpretation), planned start 2026-05-22.
- Write Final Report using Google Docs, incorporating Part I findings and, once available, Part II findings.
- Build Jupiter Notebook: Raw Data, Cleaned Data, Annual Averages, SWIFT Analysis, Correlation Table.
- Complete all chart insertions into report with proper titles, axis labels, and captions.
- Set up shared Google Drive folder and upload all deliverables.
- Package and submit final deliverable set.

#### C. Key Accomplishments

- Full Part I analytical coverage: All 30 + 6 bonus questions answered no questions skipped or deferred. Part II's 16 questions are the next scheduled step, not yet started.
- Independent data acquisition: USDG dataset (366 days) collected and integrated via CoinGecko public API with full reproducibility documentation.
- Data quality verified across all records: 0 critical errors across Brent, Gold, Silver; SWIFT structural NaN confirmed as by-design; 1 USDG duplicate resolved.
- Key market finding Brent Oil: Latest value \$102.86/barrel (1 March 2026) sits well above the 80-year (1946–2026) long-term average of \$62.71/barrel.
- Key market finding Gold: Gold's 2026 annual high reached \$5,277.89/oz (course-material context: Iran-Hormuz crisis + BRICS gold accumulation), with a 2026 annual average of \$4,881.62/oz — roughly 250× higher than the \$19.25/oz low recorded in 1915.
- Key currency finding: USD retains global payment dominance at 51.14% (April 2026), well ahead of EUR (21.30%), GBP (6.54%), JPY (3.53%) and CNY (3.10%). CNY moved up from 6th to 5th place between January and April 2026, overtaking JPY.
- Correlation confirmed: Gold–Silver correlation of 0.95 (annual averages); Brent monthly volatility (9.05%) confirmed as 2.1× higher than Gold (4.30%).

4. Schedule Performance

| Milestone                               | Planned Date       | Actual Date | Status        | Notes                                                                                 |
|-----------------------------------------|--------------------|-------------|---------------|---------------------------------------------------------------------------------------|
| Project Kickoff                         | Week of 2026-04-28 | 2026-04-28  | ✔ Complete    | Assignment analysis + data review initiated                                           |
| Data Pipeline Design                    | Week of 2026-05-05 | 2026-05-07  | ✔ Complete    | Architecture defined; Collect -> Clean -> Transform -> Analyze -> Visualize -> Report |
| Data Extraction & Cleaning              | Week of 2026-05-12 | 2026-05-15  | ✔ Complete    | All 5 datasets loaded, cleaned, and validated in Jupyter notebook                     |
| Preliminary Analysis (Part I)           | Week of 2026-05-19 | 2026-05-19  | ✔ Complete    | All 30 + 6 bonus questions answered; 11 charts generated                              |
| In-depth Analysis & Reporting (Part II) | Week of 2026-05-22 | -           | 🟡 Planned     | Not yet started; scheduled to begin in 2 days                                         |
| Final Report Submission                 | 2026-07-15         | -           | 🔴 Not Started | Pending                                                                               |

Schedule note: All phases to date (Sprints 1–2: pipeline design, data cleaning, Part I analysis) completed on or ahead of plan. Part II market research is on track to begin as planned on 2026-05-22.

## 5. Budget and Resource Utilization

| Resource                                                    | Planned (hrs) | Actual (hrs) | Variance | Status               | %   |
|-------------------------------------------------------------|---------------|--------------|----------|----------------------|-----|
| Data Engineering & Analysis (Abdulhadi Kirkici + Sagar Vig) | 30            | 24           | -6       | ✔ On track           | 80% |
| Data Researchers (Parvesh Kumar + Sameep Santosh Dokhale)   | 20            | 4            | -16      | ● Ramping up         | 20% |
| Report Writer (Rimsha Shafique)                             | 15            | 1            | -14      | ● Not started        | 7%  |
| Software Tools                                              | 0             | 0            | 0        | ✔ All free           | -   |
| Total                                                       | 65            | 29           | -36      | Sprint 2 closing out | 45% |

Tools used:

- Python 3 + Jupyter Notebook: data pipeline and all analytics
- pandas, numpy, matplotlib, seaborn, scikit-learn: analysis and visualisation libraries
- Google Docs — report drafting (collaborative, not yet started)
- Trello : task management
- WhatsApp : team communication

## 6. Quality and Technical Performance

| Quality Indicator | Details                                                                                                                                                                                                                                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Accuracy     | All five datasets verified: 0 missing values in Brent/Gold/Silver; SWIFT 388 NaN confirmed as structural (long-format design, not data error, no action taken); 1 USDG duplicate date resolved via drop_duplicates(keep="first"). No negative or zero price values in any commodity dataset.                                                              |
| Pipeline Uptime   | 100%, the project's analysis notebook executed successfully end-to-end through all Part I steps without runtime errors. All 11 charts rendered and saved correctly. A date-parsing issue (MM/DD/YYYY misread as DD/MM/YYYY by default pandas) was identified and resolved using an explicit date format specification.                                    |
| Compliance        | All data handling in line with data source requirements. CoinGecko Public API free tier terms observed: data used for academic research, attribution included, no commercial redistribution. Full data collection methodology and reproducible Python code documented for academic integrity. itation prepared for all external data sources used so far. |

7. Risks and Issues

| Risk / Issue                                                       | Status              | Mitigation Action                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Gaps — USDG launch period (Nov 2024–Apr 2025) unavailable     | Resolved / Accepted | CoinGecko free tier limits historical access to 365 days. First 6 months of USDG history not available. Documented in Data Overview section with explicit limitation note. Sufficient data (366 days) available for all analytical objectives.                                                                                                        |
| Pipeline Integration Bug - Date parsing (MM/DD/YYYY vs DD/MM/YYYY) | Resolved            | Explicit format string format='%m/%d/%Y' applied to all commodity date columns. Verified via df.Date.min() / df.Date.max() sanity checks. No data loss.                                                                                                                                                                                               |
| Part II Market Research Not Yet Started                            | Planned             | Scheduled to begin 2026-05-22 per project plan. already identified and reviewed by the team in preparation.                                                                                                                                                                                                                                           |
| Team availability variance - Uneven workload distribution          | Monitored           | Technical workload concentrated on Abdulhadi Kirkici (Co-Leader) and Sagar Vig during Part I as planned. Parvesh Kumar and Sameep Santosh Dokhale are ramping up for Part II starting 2026-05-22. Rimsha Shafique's report-writing tasks have not yet begun, which is expected at this stage. Capacity monitored via Trello board and async standups. |
| Final report coordination — Google Drive not yet set up            | Open                | Tabassum Nadavinmani to create shared Drive folder ahead of report drafting. Report will be migrated to shared Drive once folder is available and Part II is complete.                                                                                                                                                                                |

8. Variance Analysis

| Area           | Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Schedule       | All completed phases (data pipeline, cleaning, Part I analysis) delivered on or ahead of the planned timeline. Part II (Q1–Q16) has not started but remains on schedule, with a planned start date of 2026-05-22,only 2 days from this report date. No schedule slippage observed to date.                                                                                                                                                                                                                           |
| Budget (Hours) | Total hours are well under the full project plan (29 of 65 planned hours used, 45%), which is expected at this stage since Part II and report writing have not yet begun. Data Analyst hours (24 of 30) are concentrated in the completed Part I pipeline and analysis work, including the USDG pipeline build and date-format debugging. Researcher and Report Writer hours will scale up materially once Part II begins. No corrective action required, the current variance simply reflects work not yet started. |

## 9. Forecast

Part II completion forecast: Week of 22 May 2026, in line with the project plan.  
Sprint 3 (final report) forecast: To begin immediately after Part II completes, contingent on:  
1. Google Drive shared folder setup (Tabassum Nadavinmani) target before Part II completes.  
2. Part II research sessions starting 22 May 2026 (Parvesh Kumar + Sameep Santosh Dokhale), feeding directly into report Section 4.  
Deliverable readiness forecast:

| Deliverable                            | Forecast Ready Date | Owner                                                                             | Status      |
|----------------------------------------|---------------------|-----------------------------------------------------------------------------------|-------------|
| Part II Market Research (16 questions) | 2026-05-22          | Abdulhadi Kirkici (technical) + Parvesh Kumar & Sameep Santosh Dokhale (research) | Planned     |
| Final Report (Word/PDF)                | 2026-06-22          | Rimsha Shafique (primary), all team review                                        | Not Started |
| Jupiter Notebook                       | 2026-06-20          | Abdulhadi Kirkici                                                                 | In Progress |
| Full Submission Package                | 2026-06-28          | Tabassum Nadavinmani                                                              | Not Started |

## 10. Recommendations

1. Start Part II Market Research on schedule. Parvesh Kumar and Sameep Santosh Dokhale should begin the 16 market-research questions (Event Identification, Commodity Linking, Currency Linking, Business Interpretation) on 2026-05-22 as planned. Abdulhadi Kirkici to keep the notebook ready to receive and format their findings alongside the Part I analysis.
2. Prepare Google Drive and report scaffolding in parallel. Tabassum Nadavinmani should set up the shared Drive folder before Part II completes, so Rimsha Shafique can begin collaborative report drafting immediately afterward with no idle time between Part II completion and report kickoff.
3. Plan section ownership for Sprint 3 now. Once Part II is complete, assign clear section deadlines: Rimsha Shafique (Sections 1–3), Parvesh Kumar + Sameep Santosh Dokhale (Section 4, building on their own Part II research), Sagar Vig (Section 5). Deciding this now avoids delay when Part II finishes.

## Authorizations

| Team Leader                                                                 | Co-Team Leader                                                           |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Signature: _____<br><br>Name: Tabassum Nadavinmani<br><br>Date: 20 May 2026 | Signature: _____<br><br>Name: Abdulhadi Kirkici<br><br>Date: 20 May 2026 |